



A Note on the Venture Capital Industry

Small firms and new business creation have become potent forces for economic development in the United States. Prior to 1980, large firms created the majority of new jobs in the American economy. During the last two decades, however, a major structural shift occurred. Fortune 500 companies lost four million jobs. At the same time, firms with fewer than 100 employees added 16 million new jobs [Birch (1990)]. This was the first time in the twentieth century that the **shift from large to small firms occurred**, and it represented a fundamental change in the nature of growth in the American economy.

During the same twenty-year period, the rate of new firm incorporation rose dramatically. By the end of the decade, over 1.3 million new businesses were being started annually. Scherer (1991) demonstrated that during the 1980s, small firms were more innovative than large firms. He found that during the 1980s, firms with fewer than 500 employees, created 322 innovations annually per million employees while large companies contributed only 225 innovations per million employees.

This dramatic shift towards small firms makes it imperative that capital for start-up companies is available at renewable prices. This paper explores the effects of the unprecedented increase in money flowing into the venture capital sector after changes in the 1979 Employee Retirement Income Security Act's (ERISA) "prudent man" rule. Prior to 1979, pension funds were severely limited by ERISA in the amount of money they could allocate to high-risk assets, including venture capital. The 1979 change explicitly allowed fund managers to invest up to 10% of their capital in venture funds. **Pension fund commitments to venture capital rose dramatically**, increasing annual new contributions to venture capital funds from \$100-\$200 million during the 1970s to almost \$70 billion by the end of 2000.

The pattern of "feast or famine" in the venture capital industry, however, is one that continues to repeat itself. The availability of venture capital has clearly grown of the last twenty five years, but it has also gone through major increases and decreases. This cyclicity is more than just an intellectual curiosity to venture groups. Historically, a pronounced relationship has emerged between fundraising and investment performance. Often, periods of accelerated fundraising activity have preceded an alarming downturn in returns.

Perhaps not surprisingly, this pattern has spawned a deep pessimism among many venture capitalists. The industry, they lament, is inherently cyclical; all players can do is ride out the waves. Each boom in fundraising sparks too rapid growth and tempts investors to take extreme risks.

Professor Paul A. Gompers prepared this note as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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Eventually, returns suffer, which in turn triggers a decline in funds raised. Then the cycle begins all over again.

This latest surge of activity brought with it much of the criticism of the early booms in the venture capital industry. Industry pundits claimed that too many deals were being funded at valuations that were unrealistic. In the spring of 2000, the large decline in technology stocks and the subsequent bursting of the Internet bubble led to a tremendous amount of finger pointing. The issue of whether the venture capital industry will sustain itself into the next millenium is clearly an important one.

1. Start-up Financing and Venture Capital

Entrepreneurs often develop products and ideas that require substantial capital during the formative stages of their companies' life-cycles. Many entrepreneurs do not have sufficient funds to finance projects themselves, and they must therefore seek outside financing. Several alternative capital sources exist. The informal risk capital market consists of individuals known as "angels." These "angels" are wealthy business-people, doctors, lawyers, and others who are willing to take an equity stake in a fledgling company in return for money to "start-up." Wetzel (1987) estimates that 250,000 individuals are active in the informal risk capital market and invest between \$20 and \$30 billion annually. Firms that require substantial amounts of money, however, may not be able to receive sufficient capital from the "angel" network because the market is dispersed with little information sharing and the amount of invested capital tends to be small (usually less than \$100,000). Banks are an important source of start-up financing for a subset of new businesses. Companies that lack substantial tangible assets and have uncertain prospects are unlikely to receive significant bank loans, however. These firms face many years of negative earnings and are unable to make interest payments on debt obligations.

Venture capital firms will finance these high-risk, potentially high-reward projects. Venture capitalists take an equity stake in the firms they finance, sharing in both upside and downside risks. Most firms that receive venture capital financing are **unlikely candidates for alternative sources of funding**. They have few tangible assets to pledge as collateral and they produce operating losses for many years.

A common misperception is that venture capital funds only high technology companies. A substantial portion of high-tech start-ups have received venture capital, including such present-day industry giants as Sycamore, Computer, Microsoft, Amazon.com, Dell, and Genentech. Yet low-tech companies such as Staples, TCBY, and Federal Express have also received significant amounts of venture capital money. Each of these firms had a unique idea or product, and venture capital was able to help the entrepreneur exploit that opportunity.

Whether the project is in a high- or low-technology industry, venture capitalists are **active investors**. They monitor the progress of firms, sit on boards of directors, and mete out financing based on attainment of milestones. Venture capitalists retain the right to appoint key managers and remove members of the entrepreneurial team. In addition, venture capitalists provide entrepreneurs with access to consultants, investment bankers, and lawyers.

The remainder of this note is organized as follows. Section 2 provides some key facts and statistics about the venture capital industry. The discussion demonstrates the large impact that venture capital has had on the U.S. economy. A brief history of the venture capital industry is presented in Section 3. Section 4 discusses the contracts and control mechanisms that the venture capitalists use. Section 5 concludes with a brief discussion of issues that are currently important in the industry.

2. Venture Capital Trends

This section is intended to present facts and statistics about the venture capital industry. The data clearly indicates that venture capital is an industry that has grown substantially over the past 15 years and one that has had a dramatic impact on the economic landscape of this country. Various factors have contributed to this success. These factors will be discussed in Section 4.

The size of annual contributions to the venture capital industry is closely tied to the IPO market. As **Figures 1A** and **1B** show, when the IPO market is active, more money flows into the industry. This relationship is largely due to the link between returns on venture capital funds and the IPO market. When the IPO market is hot, many more venture capital-backed firms can go public increasing returns in the industry. This increases investors' interest in the industry. **Figure 1A** is in inflation-adjusted terms (constant 2000 dollars). Notice that even in inflation-adjusted dollars, the industry has seen a dramatic increase in contributions. The rise and subsequent decline in venture capital commitments is discussed in the next section.

Table 1 presents a variety of statistics related to the size, activity, and success of the venture capital industry during the period 1980-200. The focus of venture capital has also shifted over time, stirring up some interesting results in the economy. The table presents a breakdown on venture capital investments by industry and by stage of development. In Panel A, the industry pattern shows several clear trends. For example, online (Internet-related) investments were so modest that Venture Economics did not break them out until 1994. Only \$61 million was invested in the online sector in 1994. By 1999, however, online related investments received \$18.5 billion. Two other industries, communications and computer software services, both took off in the late 1990s as well. These sectors both have fortunes that were closely tied to the rise of the Internet. In the early 1980s, computer hardware and semiconductor-electronics received a large fraction of venture capital investments. Medical and health related investments surged in the late 1980s and again in the late 1990s.

These waves of venture capital activity focusing on various industries at different times are driven by two factors. On the one hand, venture capitalists are opportunistic, always trying to find the latest technological breakthrough that might have an important market. Because technological innovation proceeds at different paces and with different results in the various industries, venture capital activity can become focused on different markets at different times.

A less kind picture, however, has to do with herding by venture capitalists. Venture capitalists often look around at other investors to determine what is "hot." If a venture capitalist suddenly notices that a market segment is quite active, he might invest in those segments as well. This can lead to over-funding in certain industries while still other industries with attractive growth opportunities remain underserved. The explosive growth of Internet investing in the late 1990s is just such an example. Such over investment may serve to create short-run problems for the market.

The source of commitments to the industry has also shown substantial evolution. During the 1970s, individuals were the largest source of commitments to new venture capital funds. **Figure 2A** shows that in 1978, individuals accounted for 32% of the commitments to new funds, which totaled only \$216 million. Pension funds accounted for only 15% of the total. By 1998, **Figure 2B** shows that pension funds now supplied nearly half the capital (47%) and individuals were among the smallest contributor (11%).

The investment activity of venture funds is, however, small in absolute numbers. Notice that the number of companies receiving venture financing is not large. In the most active year, 2000, only 2,200 firms received financing from institutional venture capital for the first time. While this is significantly greater than the number that received funding in the 1970s, it is only a small fraction of the 1,000,000 or so businesses that are started each year.

Investment focus has also changed over the decade as shown in Panel B in **Table 1**. As the venture portfolio of firms has grown, the number of new company investments has declined as a percentage of total capital invested. Likewise, venture funds have moved towards later stage investments. In 1980, 40% of all investments made by venture capitalists were in seed or start-up firms. By the end of the decade, that number was only 25%. Late stage investments have grown in importance over the decades.

Of the more than 2,000 firms that went public by 2000, 1,318 were still public at the end of 2000. **Table 2** presents the industry breakdowns for those IPOs and their combined market value at the end of 2000 as well as combined sales, profits, and employment during 2000 broken down by industry. As the table shows, certain industries have a large number of companies that were still publicly traded after having received venture capital financing while other industries have had very little venture capital involvement. In the energy, environmental, financial, and transportation industries, roughly a dozen or fewer companies remaining public at the end of 2000 had received venture capital backing. In the energy, financial, and transportation sectors, venture capital-backed companies represented only about one percent of the market value, sales, and profits. These industries tend to be dominated by mature companies. The relatively low importance of venture capital in these sectors mimics the low investment rate by venture capitalists in startups in these areas and likely reflects the lower propensity for major innovations. Venture capital investors hope to capitalize on revolutionary changes in an industry. Without such revolutionary change, venture capital is unlikely to be important. What is clear from the table, however, is that the overall impact of venture capital on the U.S. economy has been quite large.

Another way to assess the overall impact of the venture capital industry is to look at the economic “weight” of venture-backed companies in the context of the larger economy. By the end of 2000, venture-backed firms that had gone public made up over 20 percent of the total number of public firms in existence in the United States at that time. And of the total market value of public firms (\$8.25 trillion), venture-backed companies came in at \$2.70 trillion—over 32 percent.

Venture-funded firms also made up over 11 percent (\$460 billion) and 13 percent (\$64.5 billion), respectively, of total sales (\$4.19 trillion) and profits (\$513 billion) of all U.S. public firms at the time. And contrary to the general perception that venture-supported companies are not profitable, after-tax profit margins for these companies hit an average of 14.0 percent—substantially higher than the average public-company profit margin of 12.3 percent. Finally, those public firms supported by venture funding employed 5.6 percent of the total public-company workforce—most of these jobs high-salary, skilled positions in the technology sector. Clearly, venture investing fuels a substantial portion of the U.S. economy.

Venture investing not only supports a substantial fraction of the U.S. economy; it also strengthens particular industries. To be sure, it has relatively little impact on industries dominated by mature companies—such as the energy, financial, environmental, and transportation industries. That’s because venture investors’ mission is to capitalize on revolutionary changes in an industry and the above sectors often have a relatively low propensity for radical innovation. But contrast those industries with highly innovative ones, and the picture looks completely different. For example, companies in the computer-software industry that received venture backing during their gestation as private firms represented more than 75 percent of the software industry’s value. In the computer-hardware industry, such firms represented 78 percent of the industry’s market value, more than 60 percent of the industry’s sales and profits, and almost half of all the industry’s employees. Venture-financed firms also play a central role in the biotechnology, computer-services, industrial-services, and semiconductor industries. All of these industries have experienced tremendous innovation and upheaval in recent years. Venture capital has helped catalyze change in these industries, providing the resources for entrepreneurs to generate substantial return from their ideas

The importance of venture capital in the initial public offering market is also critical. During the 1990s, venture capital-backed firms have accounted for about 50% of the number, total amount

raised, and total market value of all initial public offering companies. This is a dramatic impact when it is realized that only a small number of firms receive venture capital financing. In certain industries (e.g., biotechnology, computers, software, and the Internet), venture capital-backed firms account for a significantly larger fraction of the IPO firms.

Table 3 shows a typical venture capital portfolio. The five firms represent investments of a venture capital limited partnership. The investments are made in multiple rounds of financing. The staged venture capital infusions (discussed below) allow the venture capitalist to monitor the progress of the firm. The cost figures for the venture capital firm represents only its investment. Other firms would have co-invested with the firm in each of these rounds.

To get a better overall picture of the relative contributions of various parties to financing growth in venture-backed companies, **Table 4** presents the financing history of eight venture capital-backed firms that have gone public. In all cases, the firms were started with an infusion of founder's capital. Multiple venture capital rounds were necessary in all the companies except Seagate. Notice that the first round venture capital investors in Apple Computer earned more than two hundred times their invested capital when the firm went public in December of 1980. These venture capital-backed IPOs are typical of the types of returns that venture capitalists earn on the capital that they invest. Initial public offerings offer the greatest return on investment for the venture capitalist.

3. The History of Venture Capital

3.1. The Roots of Venture Capital

The ancestors of modern venture capital in the United States developed in the late 19th and early 20th century. Wealthy families began to look for ways to invest in potentially high-return, high-tech undertakings. David Lampe writes in his history of the Route 128 venture capital region:

The city's [New York] great fortunes, including those of the Vanderbilts, Whitneys, Morgans, and Rockefellers, were based on such ventures as railroads, steel, oil, and banking. Although not all investors were so well known, it was wealthy families such as these that bankrolled Boston's earliest high tech entrepreneurs. When the young Scot Alexander Graham Bell needed money in 1874 to complete his early experiments on the telephone, for example, Boston attorney Gardiner Green Hubbard and Salem leather merchant Thomas Sanders helped out, and later put up the capital to start the Bell Telephone Co. in Boston. [Lampe (1992)]

The market for risk capital remained largely unorganized and fragmented throughout the late 19th and early 20th century. The first impetus to organize investing came from wealthy Americans. In the 1930s and 1940s, members of the Rockefeller, Bessemer, and Whitney families hired professional managers to seek out investment in promising young companies.

The first modern venture capital firm was formed in 1946, when MIT president Karl Compton, Massachusetts Investors Trust chairman Merrill Griswold, Federal Reserve Bank of Boston president Ralph Flanders, and Harvard Business School professor General Georges F. Doriot started **American Research and Development (ARD)** [Lampe (1992)]. The goal of the company was to finance commercial applications of technologies that were developed during World War II.

Doriot was the heart and soul of ARD and is justifiably called the "father of venture capital." Doriot's focus was on adding value to companies, not just supplying money. Companies funded by ARD were considered to be "members of the family" [Sexton (1991)]. ARD's staff under Doriot's direction began providing industry expertise and management experience to the companies they backed in order to increase their chances of ultimate success.

Doriot served as ARD's president until it was acquired by Textron in 1972. During the course of his tenure at ARD, Doriot's vision was not one of "making money" but rather financing "noble" ideas. The first investment made by ARD in 1947 was in High Voltage Engineering Company. The firm, founded by several MIT professors, was established to develop X-ray technology in the treatment of cancer. ARD invested in the company for reasons noted by Compton's comment to Doriot:

They [High Voltage Engineering Company] probably won't ever make any money, but the ethics of the thing and the human qualities of treating cancer with X-rays are so outstanding that I'm sure it should be in your [Doriot's] portfolio. [Lampe (1992)]

When High Voltage went public in 1955, the original \$200,000 investment was worth \$1.8 million.

ARD created the standard venture capital paradigm with its highly successful investment in Digital Equipment Company (DEC) in 1957. ARD invested \$70,000 for a 77% stake in DEC. Doriot's disdain for quick profits was displayed in the displeasure he expressed with Kenneth Olsen, Digital's founder and president, the first time DEC reported a profit. Doriot was concerned that not enough money was being reinvested in research and development and that the company might suffer in the long run [Kotkin (1984)]. Over the ensuing 14 years, the investment in DEC increased in value to \$355 million. Almost half of all the money earned by ARD during its 26-year existence was earned by its investment in Digital. The concept of the "home run" in venture capital was synonymous with DEC and the term would become pervasive in the industry during the 1980s and 1990s. Everyone wanted to finance the next DEC.

In 1958, the Federal government decided to play an active role in promoting small firm development by becoming a participant in and regulator of small firm financing. The Small Business Administration was given the authority to charter new small business investment companies (SBICs). SBICs were to provide early stage financing for companies in various industries. The number of SBICs increased rapidly. By the mid-1960s, 700 SBICs controlled the majority of risk capital invested in the United States.

SBICs differed markedly from ARD. SBICs tended to provide little more than money. Most managers of SBICs had little industry expertise and could not provide entrepreneurs with information or access to industry experts. SBICs did not monitor the firms as active investors, but instead relied on the repayment of loans to evaluate the success of a project.

Problems quickly developed. In an effort to lever investments in small business, SBICs were able to borrow four government guaranteed dollars for each dollar of equity capital in the investment company. Because SBICs needed to make periodic interest payments, they chose to finance firms with debt rather than equity, as ARD had chosen to do. Had they used equity, SBICs would not have been able to service their own debt obligations. Because high-risk projects are unsuited for leveraged capital structures, the use of debt financing meant that SBICs focused on more stable industries.

A second major concern was the incentive problems inherent in government guarantees. As the recent savings and loan crisis suggests, when the managers of certain financial institutions understand that the government will bail out the depositors if things go wrong, they have little incentive to monitor their investments closely. The implicit guarantee offered by the government gave individual institutions an incentive to gamble.

The initial public offering (IPO) market of the late 1960s was extremely active, and many SBICs were able to bring a number of companies public during the boom. But the IPO "bubble" and adverse investment incentives caused by the loan guarantees led to increased investments in risky projects. The recession after the first oil embargo of 1973-1974 hit young firms particularly hard. IPO activity dropped to one-tenth its previous level and many SBIC-backed firms began losing money.

SBIC-backed companies, which were often financed with debt, could not meet interest obligations. At the same time, SBICs themselves were highly leveraged and could not meet their interest and principal repayment schedules. Many were forced to liquidate. By 1978, only 250 were still active. In 1988, SBICs accounted for just 7% of venture capital financing; they provided over 75% of the investments 25 years earlier. Companies like ARD and the firms they financed survived many recessions because they relied on equity financing of both the venture capital fund and the entrepreneurial firm.

The earliest venture capital firms were organized in the Northeast, centered both in Boston and New York. It was not until 1957 that West Coast venture capital came into existence. **Arthur Rock**, then an investment banker at Hayden, Stone & Co. in New York City, was sent to investigate a potential project in California. Rock called various individual and institutional investors to secure financing for Eugene Kleiner and a group of Shockley Laboratory employees. Rock's efforts led to Sherman Fairchild, the largest holder of IBM shares at the time, who invested \$1.5 million to form **Fairchild Semiconductor**. Four years later, Rock moved to California to form the first of two early venture capital funds in Silicon Valley. His investments in such industry leaders as Intel, Scientific Data Systems, Teledyne, and Apple have had a tremendous impact in transforming high-tech in California.

The dramatic success of ARD induced individuals to start new, private venture capital firms in the 1970s. Some of the earliest imitators were actually former members of the ARD team, including Bill Congelton, the ARD associate who initiated the DEC investment. This spawning produced private venture capital firms that carried the spirit of Doriot into new companies. **The goal was always the "home run" while the modus operandi was always "hands-on" management.** The "new" venture capitalist (unlike the manager of the SBIC) provided many services to the entrepreneur, including access to investment bankers, corporate lawyers, accountants, and industry experts.

One of the early disciples of the Doriot tradition was Don Valentine. Valentine had been marketing director at National Semiconductor and was impressed by the success of ARD. Like many other early venture capitalists, Valentine brought years of industry experience to the firms he financed. When Steven Jobs and Stephen Wozniak sought his help in 1976, Valentine knew that the two engineers needed a competent manager to head their young start-up, Apple Computer. In addition to the money he provided, Valentine cajoled A.C. Markkula, Jr., a former Intel manager, to be Apple's president [Kotkin (1984)]. Valentine's help was typical of early venture capital financing. In addition to supplying \$91,000 of capital, Valentine brought in \$600,000 more in start-up financing by syndicating the investment with other venture capitalists. Prior to its initial public offering, Apple received \$3.5 million in venture capital money. That investment grew in value to \$271 million in December 1980, when Apple went public.

3.2. The Growth of Venture Capital

Prior to the 1980s, venture capital was a cottage industry. The mild "boom" of the late 1960s gave rise to the bust of the 1970s as SBICs sank into obscurity. The annual flow of money into new venture capital funds was never much more than \$200 million, and usually substantially less. As **Figure 1A** shows, money flowing into the venture capital industry increased dramatically during the 1980s. In 1987, \$4.9 billion was committed to new venture capital funds. The trends in venture capital commitments appears to be highly correlated with the initial public offerings market.

The increase in venture capital coincided with two important legislative changes. The first was the 1978 Revenue Act, which decreased the capital gains tax from 49.5% to 28%. The second was the change in ERISA's "prudent man" rule in 1979, which explicitly allowed pension funds to invest in venture capital. While both changes may have been favorable to venture capital investment at the time, the long-term impact of the "prudent man" rule change was substantially greater than the reduction in the capital gains tax rate.

Many professionals in the venture capital industry argue that the cut in the capital gains tax spurred the increase in venture capital investing. The cut may have had a marginal effect, but the overall impact was likely quite small. Prior to 1978 the tax favorability of capital gains, i.e., the difference between the highest marginal tax rate on normal income and the tax rate on capital gains was 20.5% (the maximum marginal tax rate was 70% and the capital gains tax was 49.5%). The Tax Reform Act of 1978 lowered the capital gains tax to 28% without changing the top marginal tax rate, increasing the tax advantage of capital gains to 42%. This would give an incentive to taxable individuals to invest in venture capital because the returns from venture capital investments are realized primarily in the form of long-run capital gains. While the flow of money into venture capital funds did increase substantially following the change, much of the new funds were from tax-exempt institutions.

Capital gains policy seems to have had little impact after the 1978 reform. Subsequent changes in the capital gains tax rate have had little effect on the flow of money into venture capital. The Economic Recovery Tax Act of 1981 lowered the capital gains tax to 20% but also reduced the top marginal tax rate on income to 50% reducing the capital gains tax advantage to 30%. New commitments to venture capital increased in the following two years, however. When the capital gains tax differential was totally eliminated in the Tax Reform Act of 1986, the flow of money into venture capital increased from \$4.5 to \$4.9 billion. If the incentive to invest in venture capital was affected by capital gains tax treatment, then the money committed to venture capital should have declined after both the 1981 and the 1986 tax changes.

The primary reason that capital gains taxation has a small impact on the amount of money flowing into the venture capital industry is that up to 70% of the money flowing into new funds is from tax-exempt sources such as pension funds, endowments, trusts, and foreign companies. Changes in the tax code have no effect on this group. Corporations, which can be taxed, constitute a significant fraction of the remaining commitment to venture capital funds. Corporations invest for both pecuniary and nonpecuniary reasons. Investment in venture capital is viewed as a way to buy “a window on new technology.” Many firms may be unwilling or unable to undertake new investment in research or product development. Funding the development through venture capital may provide a way to avoid the immediate “hit” on earnings that occurs because standard accounting rules require research and development to be expensed. In addition, the corporation does not bear the entire cost of the development process spreading the risk to other limited partners. Any change in the capital gains rate is likely to have a small, second-order effect on corporate venture capital investment, because firms are looking at more than the monetary gains.

The single most important factor accounting for the increase in money flowing into the venture capital sector was the change in the 1979 amendment to ERISA’s “prudent man” rule. Prior to that date, the Employee Retirement Income Security Act of 1974 prohibited pension funds from investing substantial amounts of money in venture capital or other high-risk asset classes. The Department of Labor’s clarification of the rule explicitly allowed pension managers to invest in high-risk assets including venture capital.

The rule change opened the door to tremendous capital resources. Pension funds controlled over \$3 trillion by the end of the 1980s. Stocks and bonds performed extremely poorly during the 1970s, the same time that venture capital was earning in excess of 25% per year. Pension fund managers saw venture capital as a way to earn excessive rewards. But they did not always understand the inherent risk associated with higher returns, and they poured money into new funds at a rapid pace. **Figure 2A** shows that in 1978 when \$218 million was invested in new venture capital funds, individuals accounted for the largest share (32%). Pension funds supplied just 15%. By 1998, when more than \$17 billion was committed to new funds (**Figure 2B**), pension funds accounted for 47%, by far the largest share. The participation of individuals had fallen to among the lowest fraction of new money committed (11%).

At the same time, commitments to the venture capital industry have been very uneven very uneven. As **Figure 1A** depicts, the annual flow of money into venture funds increased by a factor of ten during the early 1980s, peaking at just under six billion (in 2000 dollars). From 1987 through 1991, however, fundraising steadily declined. This fall-back reflected the disappointment that many investors encountered with their investments. As **Figure 3** shows, returns on venture capital funds declined in the mid-1980s, apparently because of overinvestment in various industries and the entry of inexperienced venture capitalists. As investors became disappointed with returns, they committed less capital to the industry.

This pattern reversed dramatically in the 1990s, which saw rapid growth in venture fundraising. The explosion of activity in the IPO market and the exit of many inexperienced venture capitalists led to increasing venture capital returns. New capital commitments rose in response, increasing by more than twenty times between 1991 and 2000. While previous investment surges have been associated with falling venture capital returns, the most recent expansion in fundraising has seen a rise in the returns to venture funds. Much of the recent growth in fundraising was fueled by public pension funds, many of which entered venture investing for the first time in a significant way.

The institutionalization of venture capital has had some dramatic effects on the venture capital industry and its performance. Because money was poured into venture capital and the flows were directed on the basis of short run performance measures, the structure of venture capital investment deteriorated dramatically.

Academic research has investigated short-run investment behavior of money managers. The myopic horizons of pension fund managers result from quarterly evaluations of the funds. A pension fund manager cannot afford to have a poor year or she risks losing her job. Because the returns from venture capital may take five to ten years to show their results, the mismatch of time horizons could have serious consequences. *Institutional Investor* noted in 1984 that a potential problem existed:

In fact, the only thing that could destroy this market, [Raymond] Held [vice president of venture capital investing at Manufacturer Hanovers Investment Corp.] and his colleagues insist, is pension funds themselves and their obsession with the short-term relative performance game. Venture investing requires patience as problems are worked through and with all the money that has flowed into venture investments the last few years, the problems are bound to multiply. [Jansson (1984)]

3.3. Pension Funds and Venture Capital

The huge increase in institutional money had a dramatic impact on the process of venture capital investing. This process can be broken down into three main stages. All three have been adversely affected by the institutionalization of venture capital.

The first stage is identification of deals. For each 100 business plans that a typical venture capitalist might review, she will invest in only one. An experienced venture capitalist concentrates on not only the skills and ideas of the entrepreneur but also on his personal qualities.

The second stage is the actual structuring of the deal. The deal contains contractual elements including the type of financing (e.g., convertible preferred stock, common equity, etc.), timing of capital infusions, explicit and implicit options, board representation, and advice provided.

The final stage is the harvesting of investments. During this phase the investment is made liquid by performing an initial public offering, negotiating a merger, acquisition, buy-back of the entrepreneurial firm, or liquidating the assets.

The move to late-stage investment The dramatic rise in pension fund money flowing into venture capital created a mismatch between the duration of most venture capital investments and the time horizon of investors. Incipient investments in firms are known as “seed and start-up” investments. In 1980, 40% of all venture capital investing was in these classes. By 2000, seed and start-up investing accounted for only 25% of new investments, nearly half the fraction of 1980.

The shift to late-round financing is evidence that short-term performance pressures increased dramatically with the institutionalization of venture capital sources. Seed and start-up projects may take five to ten years to show returns. The industry practice is to wait to write-up the investment until it is harvested. Because valuation of venture capital-backed firms is very difficult and subject to potential biases, institutions have no idea how their portfolio is performing for several years. In fact, bad venture capital projects are usually identified much sooner than are good ones. Most bad investments are written-off early in the fund’s life, depressing the stated value of the venture capital portfolio. Because one bad year could end the career of a money manager, pension fund investors put significant pressure on venture capitalists to perform quickly. Kaplan (1991) shows that the lifetime of an LBO investment is significantly shorter than that of a comparable venture capital investment. Assets are sold off almost immediately to meet debt burden, and many companies go public again (in a reverse LBO) in a very short period of time.

The shift towards later-stage investing and LBOs is a direct result of the need to realize returns earlier. With less money flowing into the developmental stages of investment, the U.S. economy may be heading into a period in which new firms will be less innovative. The success of venture capital in the 1980s was driven by the tremendous amount of money invested during early-stage projects in the late 1970s and early 1980s. The dramatic decline in early-stage investing may mean that there will be fewer opportunities for profitable late-stage investing in the 1990s.

Herding and venture capitalists The large increase in money also affected the identification of opportunities in a perverse way. Because venture capitalists were now bloated with cash, they could pursue many projects. Certain industries appeared particularly attractive and a herd mentality resulted. **Too much money was chasing too few deals and venture capitalists paid far too much for the projects.**

Sahlman and Stevenson (1987) chronicle the exploits of venture capitalists in the Winchester disk drive industry. Sahlman and Stevenson believe that a type of market myopia affected venture capital investing in the industry. During the late 1970s and early 1980s, \$400 million was invested in 43 disk drive companies. Two thirds of this investment came between 1982 and 1984. Many disk drive companies also went public during this period. Over \$800 million was raised in public offerings by 1983. While industry growth was rapid during this period of time (sales increased from \$27 million in 1978 to \$1.3 billion in 1983), it is questionable whether the scale of investment was rational given any reasonable expectations of industry growth and future economic trends.

In mid-1983, the 12 publicly traded disk drive companies had a market value of \$5.4 billion, which represented four times sales and a price-to-earnings ratio of nearly 50. The bubble had to burst, and it did in 1984. By year’s end, the market value of the 12 public disk drive companies had fallen to \$1.4 billion. Industry income fell 98% as overproduction and competition cut margins. Many of the industry’s leading companies went bankrupt during this period, including Priam and Miniscribe.

The sudden and dramatic increase in venture capital commitments meant they had to find something in which to invest. The large fraction of inexperienced venture capitalists changed the rules of investing. Too many companies were funded in various industries at prices that were clearly unwarranted. The dramatic failure of disk drive and other venture capital-backed companies substantially reduced returns in the venture capital industry.

Grandstanding in the venture capital industry The large growth in new funds affected fund returns in additional ways. Gompers (1996) has explored the process of initial public offerings by venture capital-backed firms. The number of new venture capital partnerships increased from 225 in 1979 to 674 in 1989. Gompers finds that unseasoned venture capital firms (those that have been in existence five years or less) are under tremendous pressure to perform during the initial stages of their first fund. Young venture capitalists are concerned about having a track record and showing pension fund investors that they are financing worthy projects in order to raise additional funds. The empirical results show that these inexperienced venture capitalists have an incentive to “grandstand,” or bring firms from their first fund to the public market sooner than would otherwise be optimal. Pressure to perform early IPOs seems to have increased as pension funds became the major supplier of money for venture capital funds. The short-term focus of pension fund managers may be a contributing factor. Because two-thirds of the venture capital partnerships were founded during the past 10 years, returns in the industry have been dramatically affected by the increasing institutionalization of venture capital and its effect on young venture capitalists’ incentive to signal their abilities.

The effect on returns in the venture capital industry The above structural and behavioral changes in the venture capital industry dramatically affected returns. **Figure 3** shows the median return on all venture capital funds tabulated from the Venture Economics database. The figure demonstrates that median returns on venture capital investments increased from the early 1970s and peaked in 1982 at 31%. Subsequently, fund performance has decreased, and in 1990 median annual returns were only 1%.

The low returns in the early 1970s resulted primarily from the oil shock-induced recession and the failure of many SBICs. As the industry shifted toward more efficient limited partnerships that used equity financing and provided advice, and the economy expanded in the early 1980s, venture capital returns increased. During the 1980s, funds became swollen with money and problems arose. While the twenty plus percentage decline in venture capital returns since 1982 may not be entirely due to the factors listed above, they are certainly major contributors to the precipitous drop.

The declining commitment to new venture capital funds in the late 1980s and the early 1990s resulted primarily from pension funds losing interest in venture capital even though some of the excesses had been corrected. Many of the partnerships that performed poorly went out of business. Understanding the nature of venture capital can also change investment behavior and many venture capitalists gained considerable experience during the difficult years of the late-1980s.

The story is all too familiar. Excesses in investment lead to new excesses in pulling out. C. Kevin Landry, managing partner at TA Associates (a leading venture capital firm) notes: “Pension funds all wanted to get into venture capital in 1983, and that was the wrong time [because returns had peaked]. Now, most institutions want to get out at the wrong time.” [Retkwa (1990)]

While the surge in pension fund contributions to venture capital did dramatically reduce returns in the industry in the late 1980s, other factors were important. The returns on venture capital investments are affected by the strength of the initial public offering market. If venture capitalists cannot make their investments liquid, then they stand to lose substantial amounts of money. Alternative avenues of harvesting the venture exist and an increasing number of venture capital-backed firms have been merged or acquired by large corporations in recent years. But alternative exits provide returns on investment that are significantly less attractive than IPOs and still remain a second-best harvest. The surge in IPO activity in the 1990s once again boosted returns and venture capital contributions. As we know, many of the excesses of previous booms seemed to have played themselves out as well. The issue now is whether current investors in venture capital will once again lose their appetite for the asset class.

The supply of new technologies is also an important requirement for venture capital investing. Many technologies and companies have been spawned from large corporations as a by-

product of government-funded research. Federal funding for scientific research, both military and nonmilitary, has decreased in real terms over the last 20 years.

4. Control Mechanisms in Venture Capital Contracts

During the screening process, venture capitalists review proposed business plans for young companies and design contracts that minimize potential agency costs with entrepreneurs. [These projects are characterized by their extreme riskiness, information gaps, and potentially high rewards. Venture-backed success stories include Digital, Apple Computer, Genentech, and Federal Express. Venture Economics (1988), however, noted that 6.8% of all venture-backed projects accounted for almost 50% of the return to venture capital funds. Of all projects, 34.5% experienced either a partial or total loss of invested capital. The variance in potential outcomes is clearly quite high.] Sahlman's extensive field study (1990) describes venture capital in terms of the control mechanisms employed to manage these costs. Three control mechanisms are common to nearly all venture capital financing: 1) the use of convertible securities; 2) syndication of investment at each stage; and 3) the staging of capital infusions.

When the entrepreneur and the venture capitalist negotiate a deal, they usually sign an extensive document that contains many covenants and restrictions as well as the actual terms of the financing. Elements of the financing agreement between the venture capitalist and the entrepreneur often cover the following:

- amount and timing of investment
- form of investment (e.g., convertible preferred)
- terms of the investment (e.g., conversion price, liquidation preference, dividend rate, voting rights)
- puts and call rights
- registration rights
- preemptive rights and rights of first refusal
- option pools
- employment contracts
- vesting schedules and buy-back provisions
- information rights
- board of director representation.

In addition to structuring the investment, venture capitalists are active investors. They periodically visit the firm. They may review monthly and sometimes weekly reports. They also provide advice for the firm. This may be of help in recruiting key management or in analyzing new market opportunities. The venture capitalist also provides access to professionals like attorneys, consultants, investment bankers, and bankers. Because the venture capitalist likely has repeat relationships with these individuals, the quality of the attorney, consultant, etc. is probably much higher than the firm could acquire on its own.

4.1 Convertible Preferred Stock and Venture Capital Investments

Convertible debt and convertible preferred stock are the financial instruments that venture capitalists employ most often. Sahlman (1990) hypothesizes that the most important role of convertible securities is to properly align the incentives of entrepreneurs and provide information to venture capitalists.

A key feature of the contracts and operating procedures is that risk is shifted from the venture capitalists to the entrepreneur. The entrepreneur's response to these (contractual) terms enables the venture capitalist to make informed evaluations and judgments. . . . the convertible preferred security shifts some of the costs of poor performance to the entrepreneurial team.

The implication of Sahlman's statement is that the security acts as an incentive scheme to sort out bad entrepreneurs and limit risk-taking behavior.

Sahlman notes that staged capital infusions are the most potent control mechanism a venture capitalist can employ. Prospects for the firm are periodically reevaluated. The shorter the duration of an individual round of financing, the tighter the venture capitalist monitors the progress of the entrepreneur. The role of staged capital infusion is analogous to that of debt in highly leveraged transactions. It keeps the entrepreneur on a "tight leash" and reduces potential losses from bad decisions.

In deciding whether to continue funding a troubled project, venture capitalists recognize that the high-risk implies that the project might still be ultimately very profitable. Many venture capital-backed firms including Federal Express seemed on the verge of complete ruin only to become major success stories.

Why can other financial intermediaries (e.g., banks) not do the same sort of evaluation? First, banks cannot use equity to fund the project because **regulations limit banks' share holdings.**¹ Furthermore, banks do not have the necessary skills to evaluate projects with few collateralizable assets and significant uncertainty. In addition, Petersen and Rajan (1994) argue that banks in competitive markets will be unable to finance high risk projects because they cannot extract rents in subsequent transactions. Many venture capital partnerships specialize by industry and are in better positions to determine the value of continuing funding. Also, because the probability of failure is so high, venture capitalists need a substantial fraction of the equity in order to recoup their investment. Most banks are unlikely to lend money to high risk projects. Even if they did, the required interest payments on a bank loan for a project with these risk characteristics would be extraordinarily high and might cause liquidity problems that would severely limit the firm's growth.

Most venture capital investments are made in syndicates. One venture firm will originate the deal and will look to bring in other venture capital partnerships. This syndication of investments serves several purposes. The first is that bringing in other venture firms to do their own due diligence acts as a second option on the investment opportunity. There is usually no clear-cut answer with any of the investments the venture firm undertakes. Having other investors okay the deal makes it much less likely that bad deals get funded.

The syndication of investment also allows the venture capital firm to diversify into more investments. If the venture capitalist had to invest the entire amount into all its companies, then she could make many fewer investments. Because the total risk of any particular investment is so high

¹Banks and their affiliates in other countries do venture capital-like financing [Sahlman (1992)], but their ability to hold equity is critical.

compared to the systematic risk, diversification is very beneficial. By investing in more projects (the typical venture capital fund invests in 15 to 20 companies), the venture capitalist can largely diversify away idiosyncratic risk.

The syndication pattern of venture capitalists show sharp stratification. The high quality, reputable venture funds syndicate among themselves. Many venture firms hope to break into the syndicate. Firms that establish reputations as superior investors are more likely to be invited to participate in syndicates with more reputable players. The goal of many firms is to break into those ranks [Lerner (1994a)].

5. Issues Confronting the Venture Capital Industry

Will venture capital thrive in the 2000s? The question cannot be easily answered because it is too early to tell what policies the Bush administration will succeed in implementing to stimulate small business formation in general and venture capital in particular. Health care reform could spur investment in promising areas like biotechnology and medical-related fields. The experiences of the 1980s and the political debate of the 1990s have focused attention on small firms as the engine for economic development in the United States for the next decade. New initiatives and regulations can aid in the growth of venture capital.

First, the Federal government should increase the financing of basic science research. The potentially profitable spin-offs from such spending are enormous. Spending on space and defense research created the electronics, modern communication, and computer industries. National Science Foundation funding of genetic research led to the breakthroughs that made biotech companies possible. No one can pick the next major technological advance with perfect certainty, but basic science research is an important element of any economic proposal to foster new businesses.

The constant debate over whether a lower capital gains tax would increase venture capital and the horizons of venture capital investors does not address the entire problem. Reduction in capital gains taxes alone would likely have little or no effect on venture capital investments in the absence of other changes. While venture capitalists may argue vehemently for decreases in the capital gains tax rate, the ones who would benefit most from such a reduction are the venture capitalists themselves. Because a significant fraction of venture capitalists' compensation is taken as a percentage of the accrued capital gains, reduction in capital gains tax would substantially increase their personal returns.

Publicly traded venture capital companies are a possible solution or should at least be seen as a complement to private partnerships. ARD was a public company and periodically issued equity to raise new capital. Incentives for long-term investment could be achieved because venture capitalists could reinvest proceeds as retained earnings. In addition, individuals might not be as myopic as institutional investors. Regulations limit the appeal of this option, however. Under the Investment Company Act of 1940, public venture capital firms were restricted from transactions with their portfolio companies and investors [Huemer (1992)]. The Small Business Investment Act of 1980 eased some of the restrictions by allowing public venture capital firms to incorporate as Business Development Companies (BDC) that could invest in much the same manner as limited partnerships.

The double taxation of corporate profits remains a problem. In order to avoid the extra tax burden, the venture capitalist has two options. The first is to register as a BDC with the SEC. The registration must be renewed annually and is very expensive and burdensome [Huemer (1992)]. The firm can also incorporate as a public partnership. This option limits the firm because the public partnerships are highly illiquid. No organized market exists to trade their shares. Broker and underwriter fees can also be substantial [Huemer (1992)].

In order for viable public venture capital firms to be a potential alternative to the present limited partnership form, multiple legislative changes must occur. First, public venture capital firms must be exempted from the double taxation of corporate profits. Second, filing requirements need to be simplified. Finally, if the appropriate measures are taken, a cut in the capital gains tax may increase funds available to venture capitalists.

Given the excesses of the late 1990s, one might ask what the proper scale of the venture capital industry is. Was the industry too large? The answer is both yes and no. Many venture capitalists that received money in the “boom” of the 1990s had little or no previous industry experience. These firms did not understand the nature of venture capital investing, the optimal deal structure, and effective monitoring. The industry shake-out of that will occur over the next several years will reposition venture capital for steady growth, however. Entrepreneurs exist who need financing. Venture capital investing accounts for only about 1% of capital expenditure annually. Experienced venture capitalists are and will be in short supply for some time. As long as the supply of funds does not outstrip the managerial capabilities in place, the venture capital industry can grow and be a large part of the economic landscape.

The importance of small business in the any economy makes venture capital a central part of any future economic growth. The importance of long-term perspectives is not to be underestimated. Many projects take decades to show their full benefit. If capital suppliers do not have equally long horizons, the process of new firm development, effective product development, and cutting-edge research will be hindered.

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Table 1 Selected Data on the United States Venture Capital Industry 1980-2000

Panel A: Dollars Invested by Industry (in millions).

Industry	Year										
	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Online Specific	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Communications	70.8	165.7	224.7	458.3	474.9	548.6	548.9	469.3	895.7	720.8	421.4
Computers Software and Services	18.0	50.5	153.7	366.0	452.4	412.5	464.9	472.0	432.8	482.1	615.6
Other Products	83.0	122.2	122.8	197.2	211.8	251.0	563.3	530.0	904.0	789.5	468.0
Medical and Health Related	49.5	103.5	113.9	271.3	310.0	344.8	363.8	520.8	552.8	756.4	497.0
Semiconductor and Other Electronic	74.2	163.0	202.2	329.4	444.5	437.5	477.4	451.6	433.8	331.2	264.6
Consumer Related	52.3	51.5	131.4	280.2	249.4	254.0	517.9	764.5	808.9	865.1	435.0
Computer Hardware	152.2	361.3	608.4	1,169.8	995.2	747.2	769.2	580.8	534.8	428.9	277.1
Biotechnology	48.9	100.0	83.9	144.6	116.2	141.4	271.6	284.2	326.1	332.1	284.3
Industrial and Energy	152.7	272.6	250.8	265.2	326.4	458.8	314.8	353.8	305.8	447.3	227.0
Total	701.6	1,390.3	1,891.8	3,482.0	3,580.8	3,595.8	4,291.8	4,427.0	5,194.7	5,153.4	3,490.0

Panel A: (continued)

	Year									
Industry	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Online Specific	0.0	0.0	0.0	60.9	242.5	778.8	1,440.0	3,284.7	18,513.1	25,246.1
Communications	295.8	1,107.2	694.2	957.4	1,081.2	1,669.0	2,396.4	3,318.5	8,335.4	17,627.8
Computers Software and Services	459.2	646.2	1,370.9	722.0	801.9	1,783.2	2,698.6	3,834.7	7,500.9	14,374.3
Other Products	284.4	778.7	524.2	420.9	575.8	1,320.3	1,406.8	2,443.7	4,551.9	6,279.4
Medical and Health Related	308.3	812.1	609.9	837.5	823.6	1,085.0	2,236.7	2,392.2	2,457.0	3,613.7
Semiconductor and Other Electronic	210.1	231.0	183.1	247.5	340.7	463.4	699.9	827.0	1,740.2	6,098.8
Consumer Related	350.0	359.4	704.4	748.9	774.3	1,061.1	1,236.0	1,083.7	1,710.4	1,665.6
Computer Hardware	232.4	251.1	150.9	252.5	345.9	351.8	427.9	556.2	1,303.8	2,279.2
Biotechnology	272.8	503.9	441.8	423.9	396.7	1,013.0	949.6	1,030.1	1,182.2	2,763.8
Industrial and Energy	153.2	162.7	175.8	192.7	338.1	423.4	551.8	441.1	751.1	1,423.8
Total	2,566.2	4,852.3	4,855.2	4,864.2	5,720.7	9,949.0	14,043.7	19,211.9	48,046.0	81,372.5

Table 1 (continued)

Panel B: Dollars invested by stage of development (in millions).

Stage	Year										
	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Early	286.7	643.4	650.1	1,286.0	1,356.8	1,045.7	1,393.3	1,339.8	1,412.8	1,307.7	1,092.9
Expansion	197.2	417.4	654.9	1,290.5	1,311.3	1,302.5	1,326.8	1,570.2	1,757.7	1,760.7	1,525.2
Later	55.6	107.8	255.0	436.0	474.2	622.1	583.5	631.6	452.8	558.9	416.1
Buyout	162.3	221.6	332.0	469.5	438.4	625.5	988.0	885.6	1,571.4	1,526.2	455.8

Panel B: (Continued)

Stage	Year									
	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
Early	770.9	1,106.8	2,070.1	1,493.5	2,167.1	3,150.2	3,502.3	5,278.7	10,776.5	18,634.3
Expansion	1,118.8	1,848.6	1,527.8	1,292.0	1,981.6	3,620.9	6,024.7	7,985.8	26,380.8	44,266.6
Later	462.8	1,137.1	783.5	1,235.0	980.0	2,117.0	2,801.3	3,661.8	8,764.0	16,274.5
Buyout	213.6	759.8	473.7	843.6	592.1	1,060.9	1,715.5	2,285.5	2,117.7	1,871.6

Source: Compiled from various issues of the *Venture Capital Journal*.

Table 2 Impact of Venture Capital-backed Companies through December 31, 2000.

Industry	# of VC Public Firms	Market Value of VC Public Firms (\$ millions)	Total Sales of VC Public Firms (\$ millions)	Total Profits of VC Public Firms (\$ millions)	Total Employment of VC Public Firms (millions)
Aerospace	1	\$215.67	\$33.27	\$6.75	4.36
Biotech	169	\$265,353.41	\$13,297.39	\$2,620.13	54.21
Communications	137	\$185,206.08	\$24,286.44	\$8,668.73	111.97
Computer Hardware	81	\$820,933.38	\$135,980.12	\$9,291.32	280.25
Computer Software	353	\$746,238.89	\$62,358.88	\$12,026.97	102.56
Consumer Goods	36	\$14,578.61	\$18,724.64	\$3,593.42	89.70
Consumer Services	89	\$32,109.70	\$49,920.55	(\$2,886.08)	50.32
Electronics	56	\$29,215.44	\$9,564.50	\$2,543.23	34.17
Energy	4	\$2,352.83	\$629.71	\$235.92	0.27
Environmental	5	\$615.73	\$403.91	\$104.15	4.14
Financial	14	\$18,877.31	\$8,871.37	\$1,352.03	11.18
Industrial Goods	55	\$19,992.51	\$11,267.91	\$2,258.13	49.54
Industrial Services	78	\$25,694.72	\$17,821.03	\$2,420.82	25.54
Medical Devices	76	\$6,784.21	\$4,462.34	\$1,993.06	28.86
Medical Services	27	\$21,724.07	\$16,304.12	\$2,670.18	274.97
Other	13	\$131,312.00	\$9,298.72	\$796.30	63.98
Semiconductors	64	\$352,026.81	\$67,791.95	\$15,817.08	207.51
Transportation	6	\$557.33	\$1,976.08	\$310.91	6.89
Computer Services	54	\$29,248.91	\$7,662.28	\$712.87	17.69
Total	1,318	\$2,703,037.61	\$460,655.21	\$64,535.92	1,418.11

Source: Compiled from Compustat and Securities Data Corporation

Table 3 Participation in Multiple Financing Rounds by a Venture Capital Fund

Company/ Date of Purchase	Security Purchased	Price Per Share	Number of Shares	Total Cost
Company 1				
5/1/85	Convertible Preferred Series B	\$0.68	525,145	\$357,099
8/1/85	Convertible Preferred Series B	\$0.68	972,531	\$661,321
3/1/86	Convertible Preferred Series C	\$2.25	444,445	\$1,000,001
4/1/87	Convertible Preferred Series D	\$4.50	66,667	\$300,002
	Total (Average Price per Share)	\$1.15	2,008,788	\$2,310,106
7/24/90	Estimated Value	\$23.00		\$46,202,124
Company 2				
6/1/85	Convertible Preferred Series A	\$15.00	20,833	\$312,495
11/1/85	Convertible Preferred Series A	\$15.00	20,833	\$312,495
4/1/86	Convertible Preferred Series A	\$15.00	25,000	\$375,000
5/2/88	Convertible Preferred Series B	\$8.60	28,588	\$245,857
	Total (Average Price per Share)	\$13.08	95,254	\$1,245,922
7/24/90	Estimated Value	\$3.27		\$311,481
Company 3				
2/1/87	Convertible Preferred Series B	\$1.15	347,827	\$400,001
7/1/87	Convertible Preferred Series C	\$1.90	131,579	\$250,000
3/15/88	Convertible Preferred Series D	\$1.60	283,326	\$453,322
	Total (Average Price per Share)	\$1.45	762,732	\$1,103,323
7/24/90	Estimated Value	\$1.45		\$1,103,323
Company 4				
2/1/86	Convertible Preferred Series B	\$0.95	1,473,684	\$1,400,000
12/1/86	Convertible Preferred Series D	\$1.85	461,808	\$854,345
7/22/87	Convertible Preferred Series D	\$1.85	141,829	\$262,384
	Total (Average Price per Share)	\$1.21	2,077,321	\$2,516,728
7/24/90	Estimated Value	\$4.34		\$9,015,573
Company 5				
11/1/85	Convertible Preferred Series A	\$0.34	1,470,588	\$500,000
3/1/86	Convertible Preferred Series B	\$0.45	2,083,333	\$937,500
3/1/87	Convertible Preferred Series C	\$0.75	1,333,333	\$1,000,000
	Total (Average Price per Share)	\$0.50	4,887,254	\$2,437,500
7/24/90	Estimated Value	\$0.00		\$0
Total Cost of 5 Companies				\$9,613,579
Total Estimate Value of 5 Companies				\$56,632,500
Total Gain for 5 Companies				\$47,018,921

Source: Compiled from an interim report to limited partners of a venture capital fund with more than \$20 million in capital.

Note: The amounts listed do not include investments made by others at the same time or at other times.

Table 4 Multiple Financing Rounds for Selected Venture Capital-Backed Initial Public Offerings

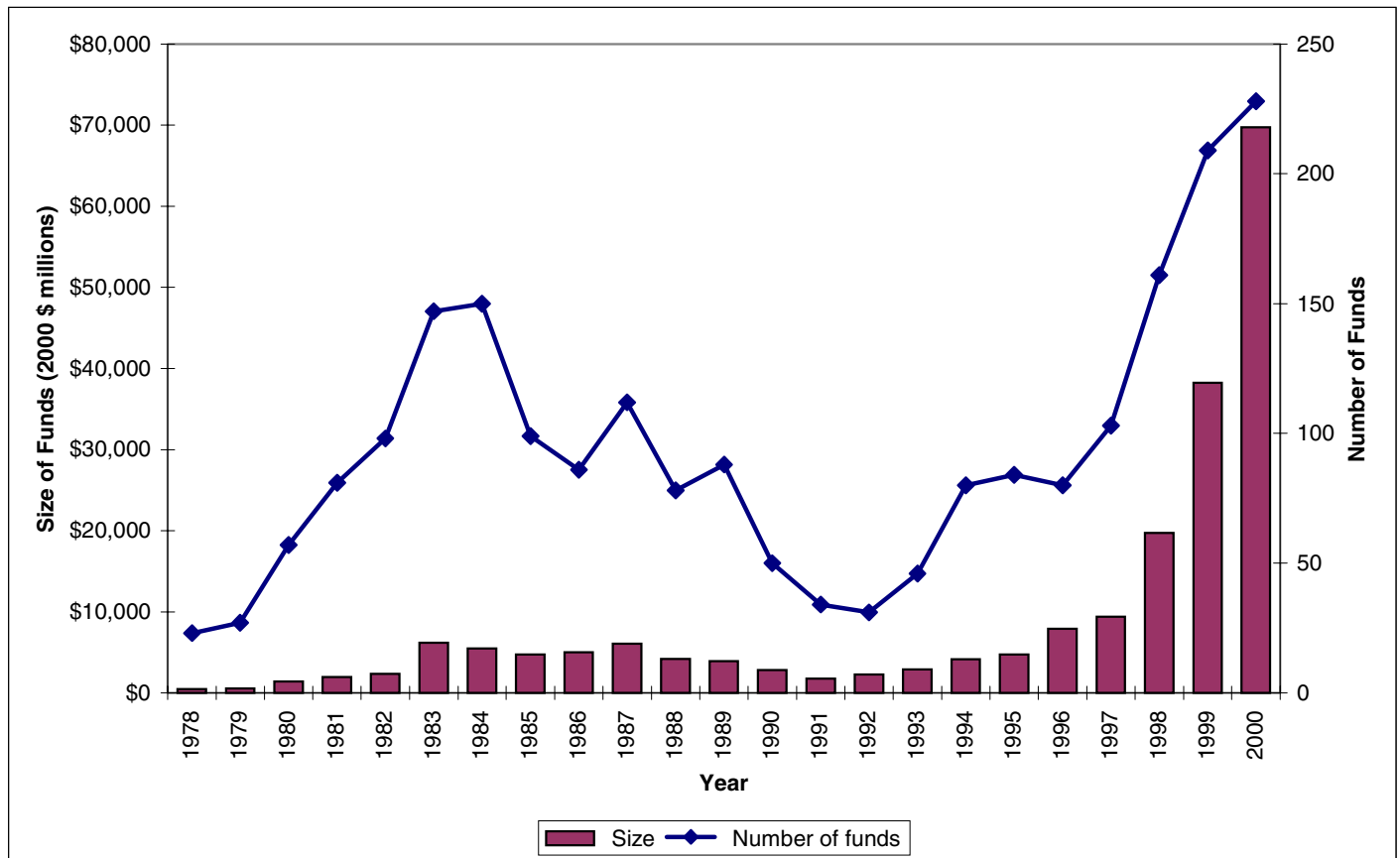
Company	Investor	Date	Amount Raised (\$,000)	Cumulative Funding (\$,000)	Stock Received (,000)	Total Shares Outstanding (,000)	Percent Ownership Acquired	Fully Diluted Valuation (\$,,000)	Price per Share (\$)	Estimated Ending Ownership
Apple Computer	Founders	Mar-77	\$1	\$1	16,640	16,640	100.0%	\$1	\$0.00	30.7%
	Founders	Nov-77	\$115	\$116	10,480	27,120	38.6%	\$298	\$0.01	19.3%
	Venture 1	Jan-78	\$518	\$634	5,520	32,640	16.9%	\$3,063	\$0.09	10.2%
	Founders	Jul-78	\$426	\$1,060	4,736	37,376	12.7%	\$3,362	\$0.09	8.7%
	Venture 2	Sep-78	\$704	\$1,764	2,503	39,879	6.3%	\$11,216	\$0.28	4.6%
	Venture 3	Dec-80	\$2,331	\$4,095	2,400	43,306	5.5%	\$42,061	\$0.97	4.4%
	IPO	Dec-80	\$101,200	\$105,295	4,600	54,215	8.5%	\$1,192,730	\$22.00	8.5%
Cray Research	Founders	Aug-72	\$2,550	\$2,550	2,869	2,869	100.0%	\$2,483	\$0.89	24.4%
	Venture 1	Jan-74	\$2,675	\$5,225	2,006	4,875	41.1%	\$6,501	\$1.33	17.0%
	Venture 2	Jan-75	\$642	\$5,867	387	5,302	7.3%	\$8,796	\$1.66	3.3%
	Venture 3	Apr-75	\$2,720	\$8,587	1,530	6,832	22.4%	\$12,146	\$1.78	13.0%
	IPO	Mar-76	\$10,890	\$19,477	4,950	11,782	42.0%	\$25,923	\$2.20	42.0%
Federal Express	Founders	Jan-72	\$4,745	\$4,745	100	100	100.0%	\$4,745	\$47.45	0.7%
	Venture 1	Sep-73	\$12,250	\$16,995	60	160	37.5%	\$32,667	\$204.17	0.4%
	Venture 2	Mar-74	\$6,400	\$23,395	872	1,032	84.5%	\$7,574	\$7.34	6.4%
	Venture 3	Sep-74	\$3,876	\$27,271	6,200	7,232	85.7%	\$4,521	\$0.63	45.8%
	IPO	Apr-78	\$25,800	\$53,071	4,300	13,535	31.8%	\$81,210	\$6.00	31.8%
Genentech	Founders	Jan-76	\$126	\$126	3,200	3,200	100.0%	\$126	\$0.04	41.4%
	Venture 1	Apr-76	\$850	\$976	1,180	4,380	27.6%	\$3,083	\$0.07	15.3%
	Venture 2	May-78	\$950	\$1,926	475	4,945	9.6%	\$9,890	\$2.00	6.1%
	Corporate	Sep-79	\$10,000	\$11,926	1,000	6,348	15.8%	\$63,480	\$10.00	12.9%
	IPO	Oct-80	\$38,500	\$50,426	1,100	7,724	14.2%	\$270,340	\$35.00	14.2%
Lotus Development	Founders	Apr-82	\$13	\$13	4,410	4,410	100.0%	\$13	\$0.00	30.9%
	Venture 1	Apr-82	\$1,000	\$1,013	3,500	7,910	44.2%	\$2,260	\$0.03	24.5%
	Venture 2	Dec-82	\$3,755	\$4,768	3,767	11,677	32.3%	\$12,044	\$1.03	26.4%
	IPO	Oct-83	\$46,800	\$51,568	2,600	14,277	18.2%	\$256,988	\$18.00	18.2%
Midway Airlines	Founders	Jun-76	\$7	\$7	700	700	100.0%	\$7	\$0.01	18.8%
	Venture 1	Jul-79	\$5,739	\$5,746	1,380	2,080	66.3%	\$8,650	\$4.16	37.1%
	Venture 2	Sep-80	\$6,000	\$11,746	789	2,869	28.3%	\$23,620	\$7.60	21.2%
	IPO	Dec-80	\$11,475	\$23,221	850	3,719	23.4%	\$53,433	\$13.50	22.9%

Table 4 (continued)

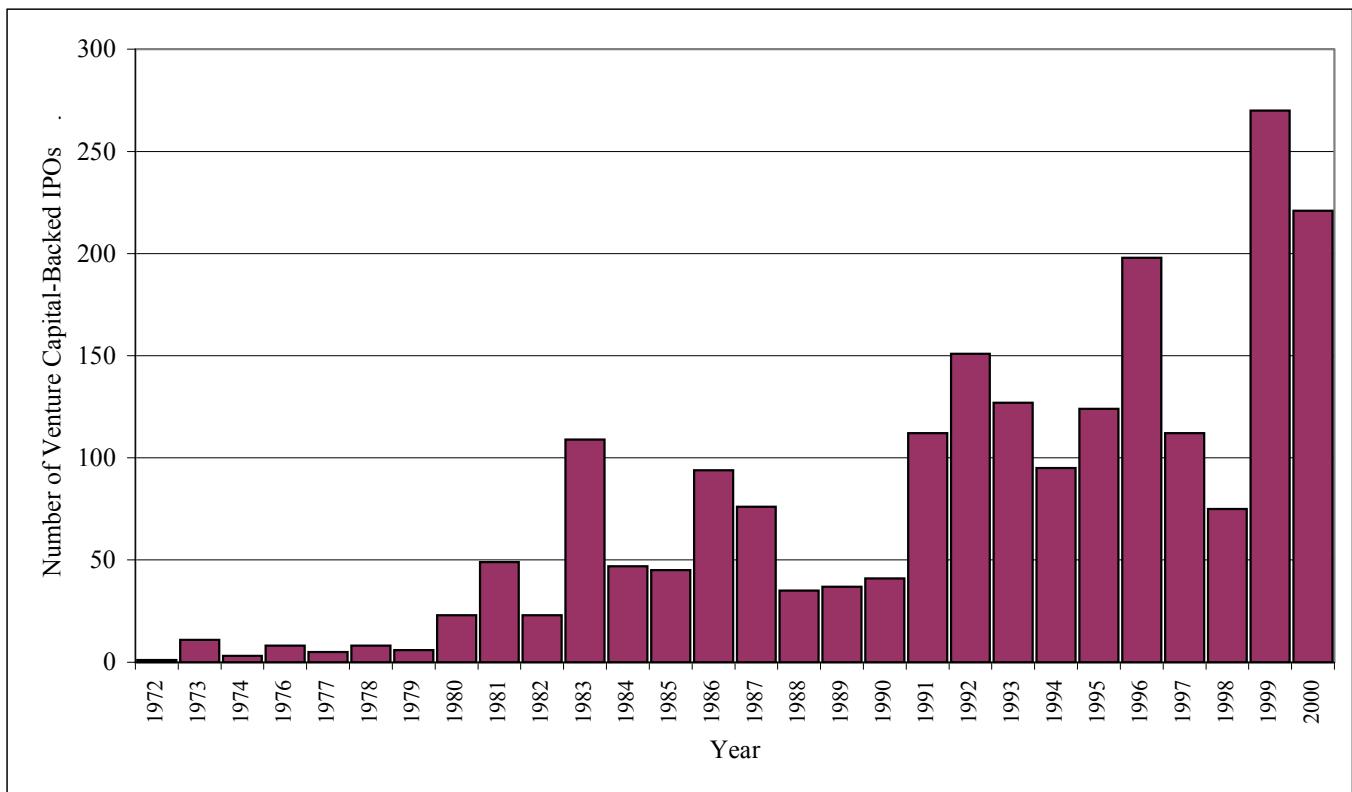
Company	Investor	Date	Amount Raised (\$,000)	Cumulative Funding (\$,000)	Stock Received (,000)	Total Shares Outstanding (,000)	Percent Ownership Acquired	Fully Diluted Valuation (\$,,000)	Price per Share (\$)	Estimated Ending Ownership
Seagate	Founders	Oct-79	\$161	\$161	11,723	11,723	100.0%	\$161	\$0.01	64.1%
	Venture 1	Jun-80	\$1,000	\$1,161	3,125	14,848	21.0	\$4,751	\$0.32	17.1%
	IPO	Sep-81	\$25,000	\$26,161	2,500	18,277	13.7%	\$182,770	\$10.00	13.7%
Staples	Founders	Jan-86	\$4,425	\$4,425	1,844	1,844	100.0%	\$4,425	\$2.40	20.2%
	Venture 1	Jan-87	\$13,927	\$18,352	2,211	4,055	54.5%	\$25,543	\$6.30	24.2%
	Venture 2	Dec-87	\$13,597	\$31,949	1,563	5,618	27.8%	\$4,871	\$8.70	17.1%
	Venture 3	Sep-88	\$2,800	\$34,749	267	5,885	4.5%	\$16,782	\$10.50	2.9%
	IPO	Apr-89	\$61,750	\$96,499	3,250	9,135	35.6%	\$173,546	\$19.00	35.6%

Source: Compiled from Annual Reports and Prospectuses.

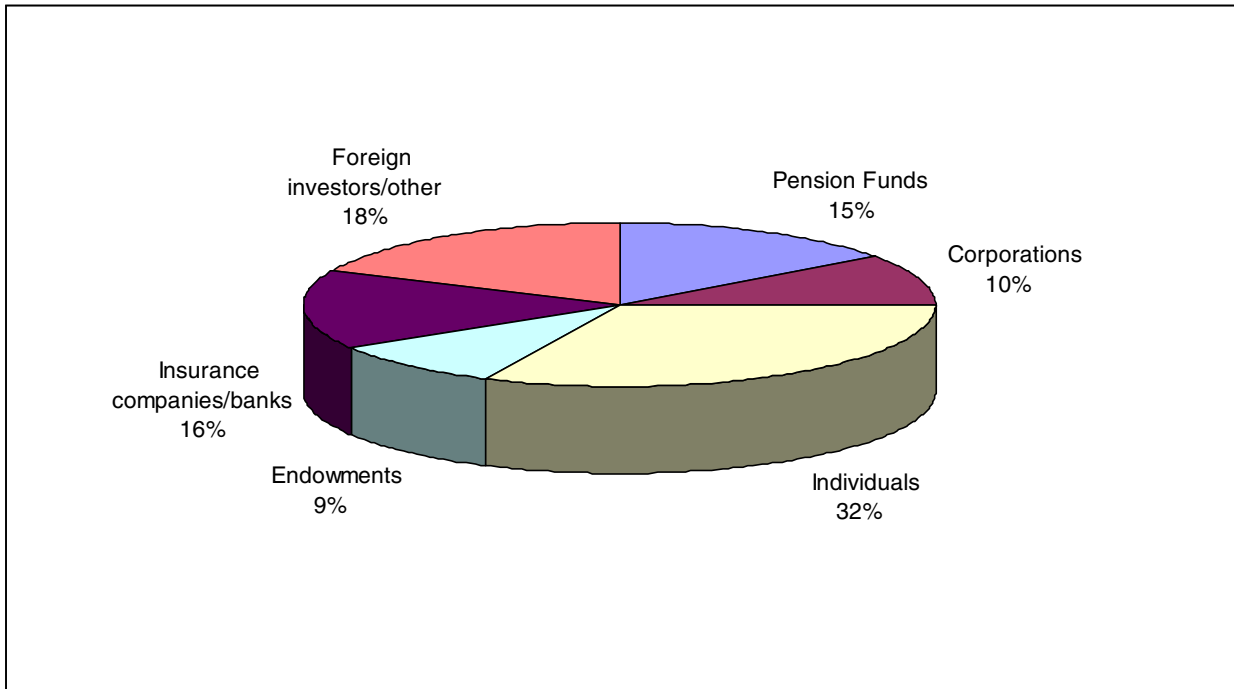
Notes:
 Venture 1, etc., represents rounds of financing from venture capitalists.
 IPO = Initial Public Offering

Figure 1A New Commitments to Venture Capital Funds in Constant 2000 Dollar

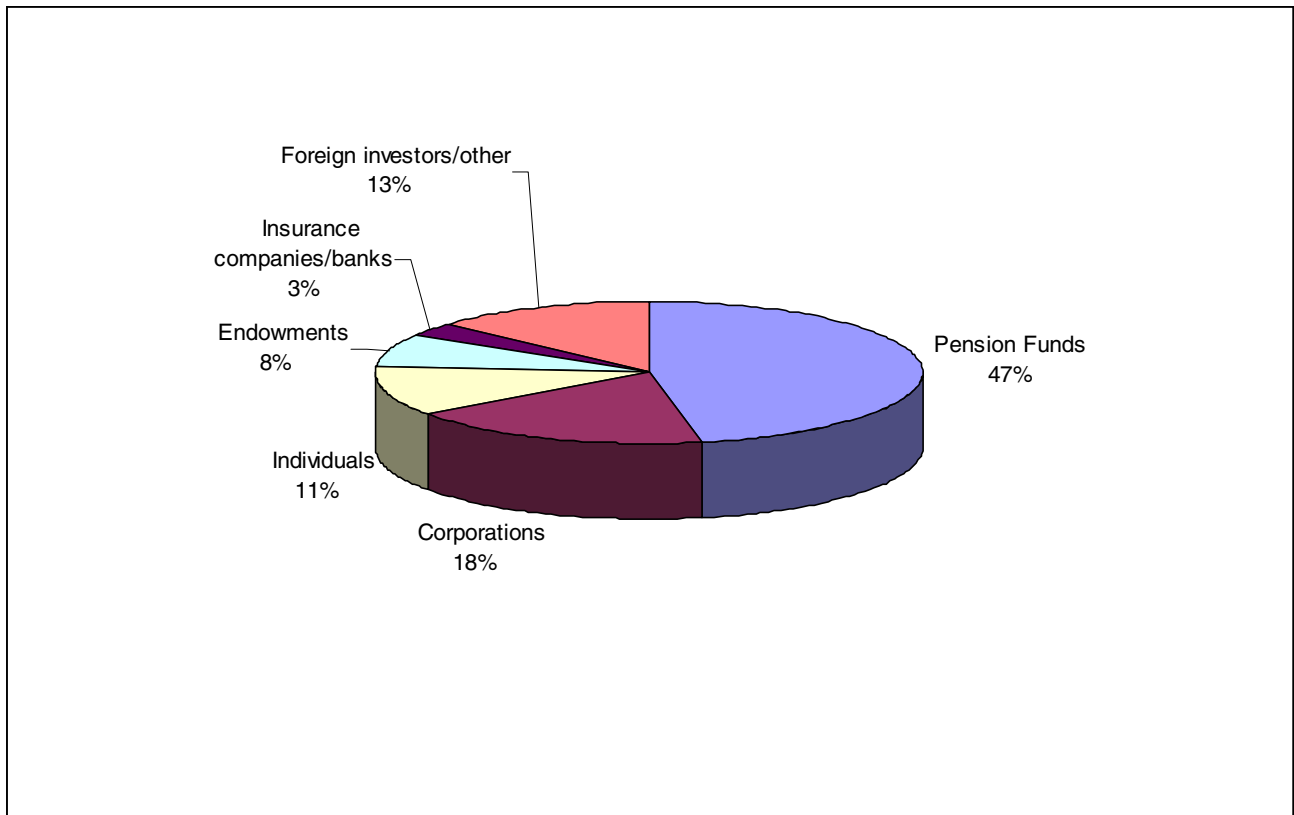
Source: Compiled from various publications from Venture Economics.

Figure 1B Number of Initial Public Offerings

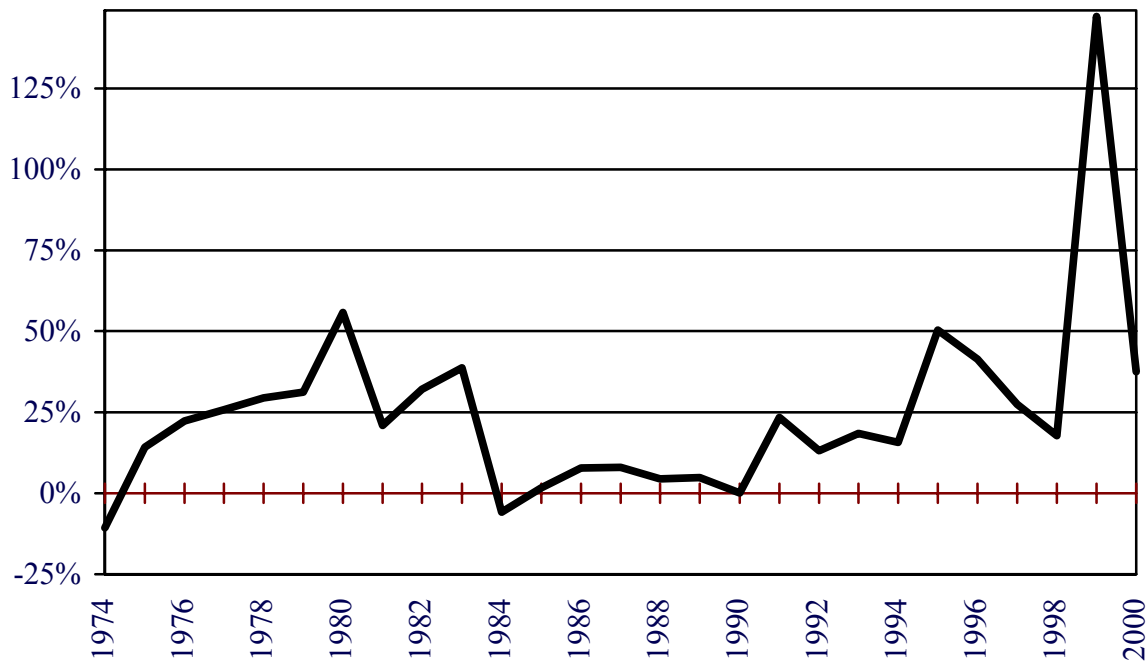
Source: Compiled from Securities Data Corporation.

Figure 2A Sources of Commitments to Venture Capital Funds 1987 - \$216 million

Source: Compiled from various publications from Venture Economics.

Figure 2B Sources of Commitments to Venture Capital Funds 1998 - \$19.7 Billion

Source: Compiled from various publications from Venture Economics.

Figure 3 Median Rates of Return on Venture Capital

Source: Compiled from various publications of Venture Economics.